6.

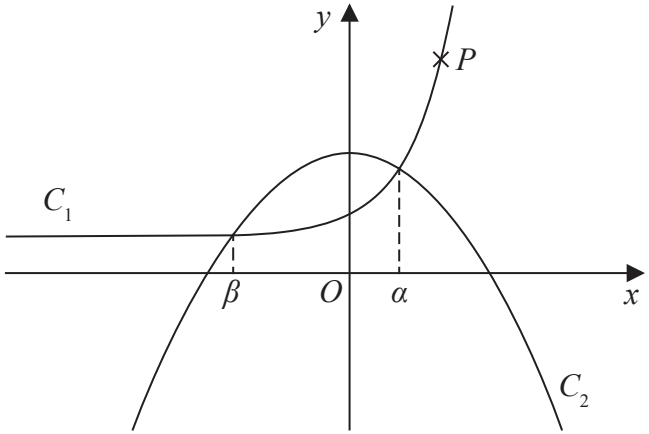



**Figure 3**

Figure 3 shows a sketch of curve $C_1$ with equation $y = 5e^{x-1} + 3$

and curve $C_2$ with equation $y = 10 - x^2$

The point $P$ lies on $C_1$ and has $y$ coordinate 18

(a) Find the $x$ coordinate of $P$, writing your answer in the form $\ln k$, where $k$ is a constant to be found.

**(3)**

The curve $C_1$ meets the curve $C_2$ at $x = \alpha$ and at $x = \beta$, as shown in Figure 3.

(b) Using a suitable interval and a suitable function that should be stated, show that to 3 decimal places $\alpha = 1.134$

**(3)**

The iterative equation

$$x_{n+1} = -\sqrt{7 - 5e^{x_n - 1}}$$

is used to find an approximation to $\beta$.

Using this iterative formula with $x_1 = -3$

(c) find the value of $x_2$ and the value of $\beta$, giving each answer to 6 decimal places.

**(3)**